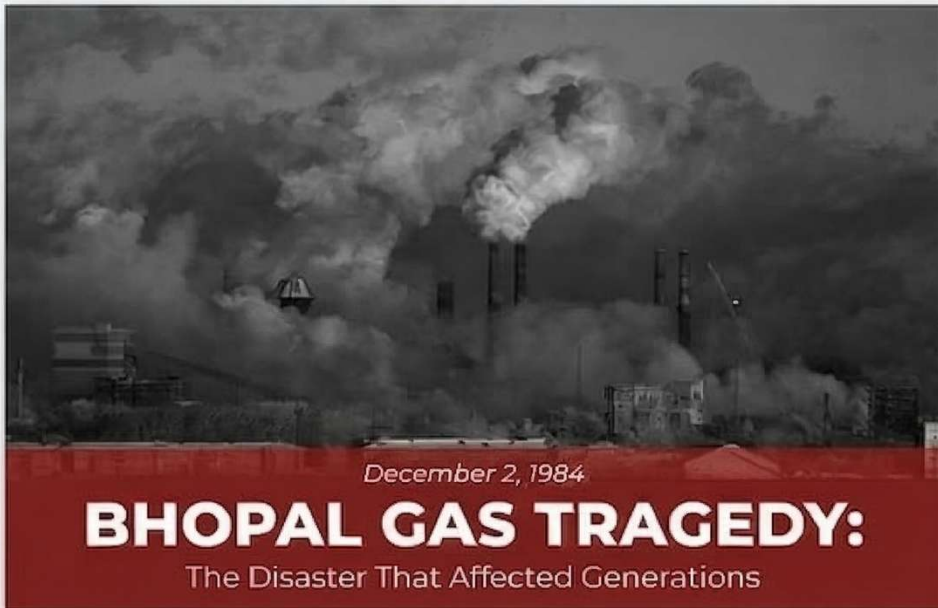




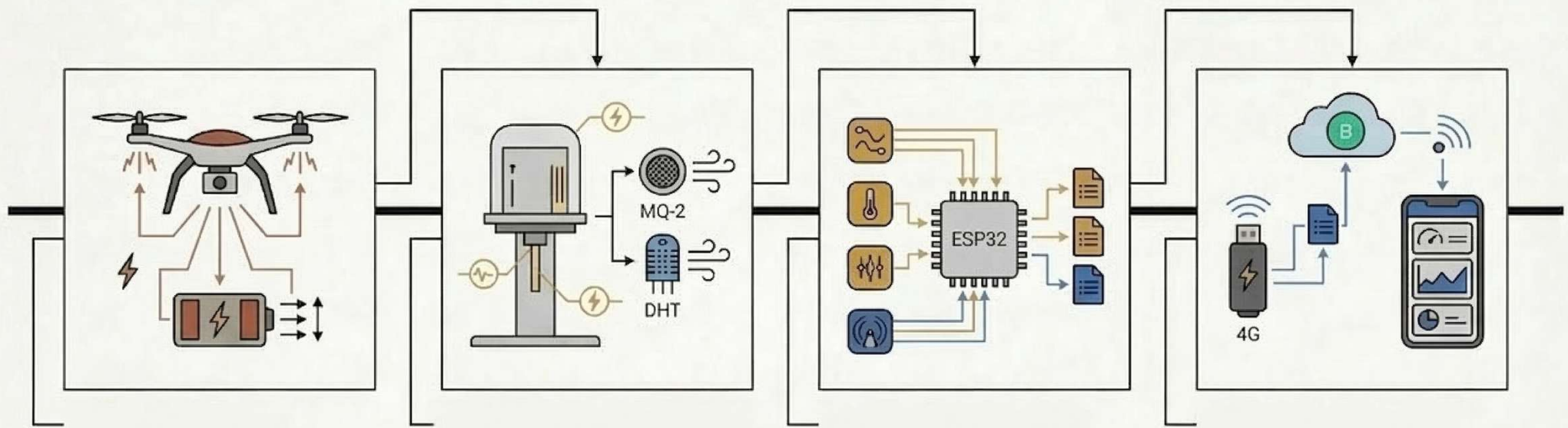
DISASTER MANAGEMENT UAV

PROBLEM STATEMENT:

DISASTER MANAGEMENT DRONES TO
PREVENT CATASTROPHIC DISASTERS

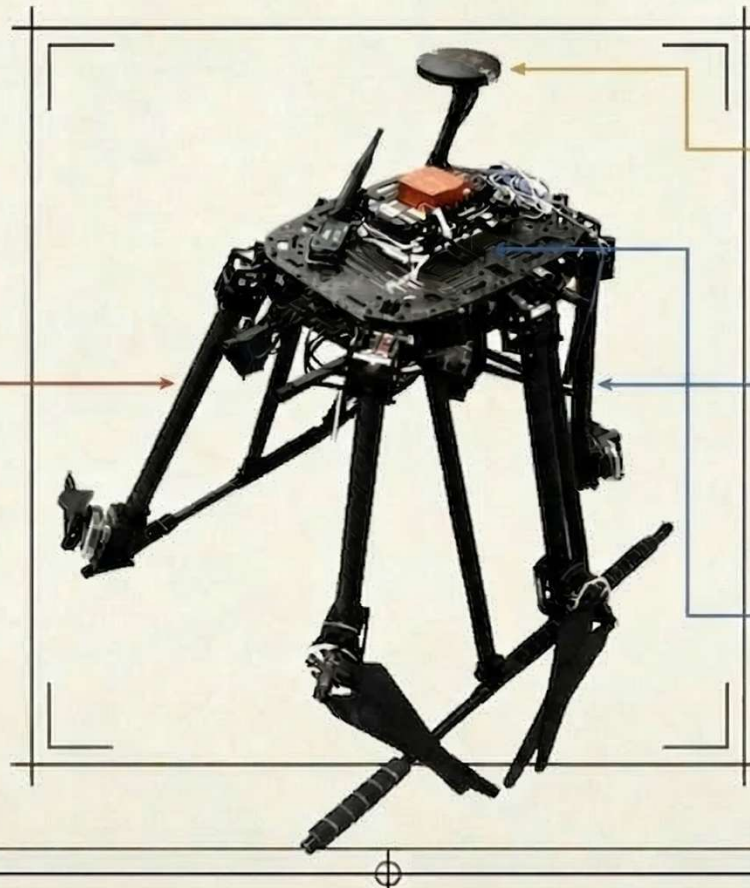


Sequence of Operations



Hexacopter UAV breakdown

Body Frame Material
Carbon Fiber Plate &
Tube Construction,
High-Strength Nylon
Arms



GPS & Sensors Module
UBLOX NEO-M8N GPS,
Magnetometer, &
Secondary IMU

**Pixhawk Flight
Controller**
Pixhawk 4 (FMUv5) with
Integrated Barometer &
Dual IMU, 4G Telemetry
& Remote Monitoring

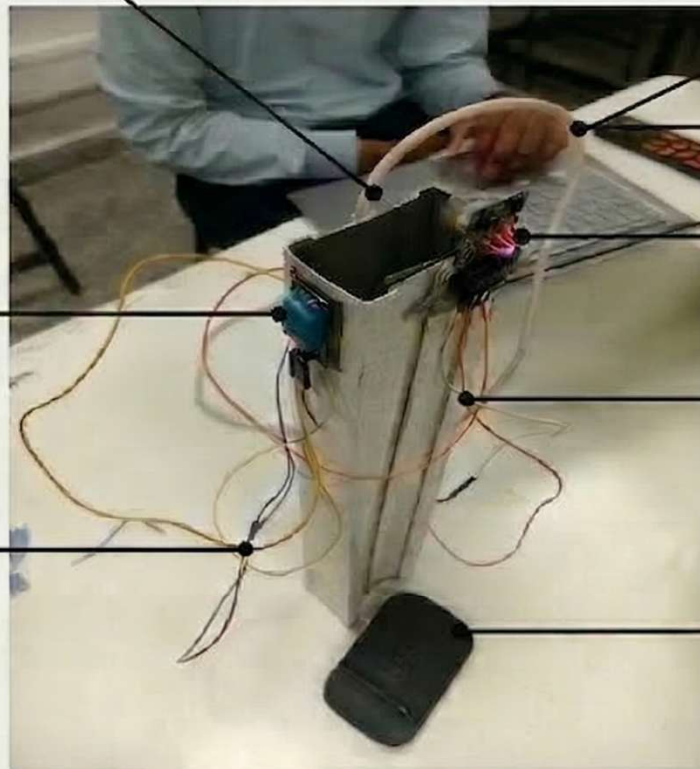
More component breakdown



ESP32 IoT Development Node
(located on internal/opposite
face, not directly visible)
(USB-B Cable connection
point visible)

DHT11 Humidity &
Temperature Sensor

Shielded Data &
Power Wire Harness



Opposite Face Node Connection
(Visible USB-B Point)
(USB-B Cable connection point visible)

MQ2 Gas & Air Quality Sensor
(Combustible/Smoke)

Custom-extruded Aluminum
Isolation Mast & Vertical Mount

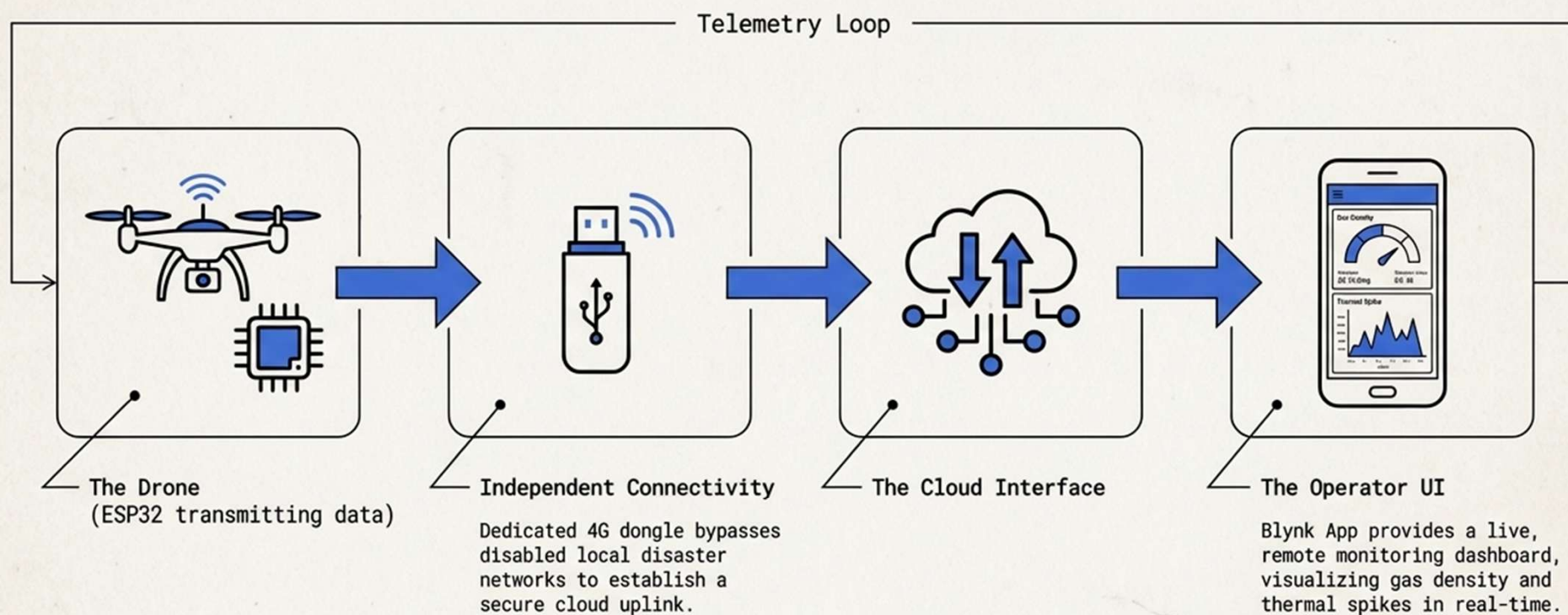
4G LTE Telemetry Dongle
& Data Link

Centralized power regulation and flight control via Pixhawk.



The Pixhawk acts as the electrical heart, actively stepping down and regulating power to prevent voltage spikes from frying the sensitive MQ-2 and DHT sensors during aggressive flight maneuvers.

The continuous remote monitoring architecture.



UI darshit

Code & Cause analysis

Blynk Cloud Integration:

The #defines directives in unique Template ID is TMPL3vvu42Yu". #define Blink Project Name is "DRP PROJECT", the Auth Token is updated, visible link to the physical ESP32 hardware to the dedicated Blynk IoT cloud dashboard.

WiFi Connectivity:

Lines 10-12, WiFi SSID (e.g. "Airtel-MyWiFi-AMF-311MN-98FA"), and urred password, e.g. ***** to establish internet connection and communicate with the Blynk server.

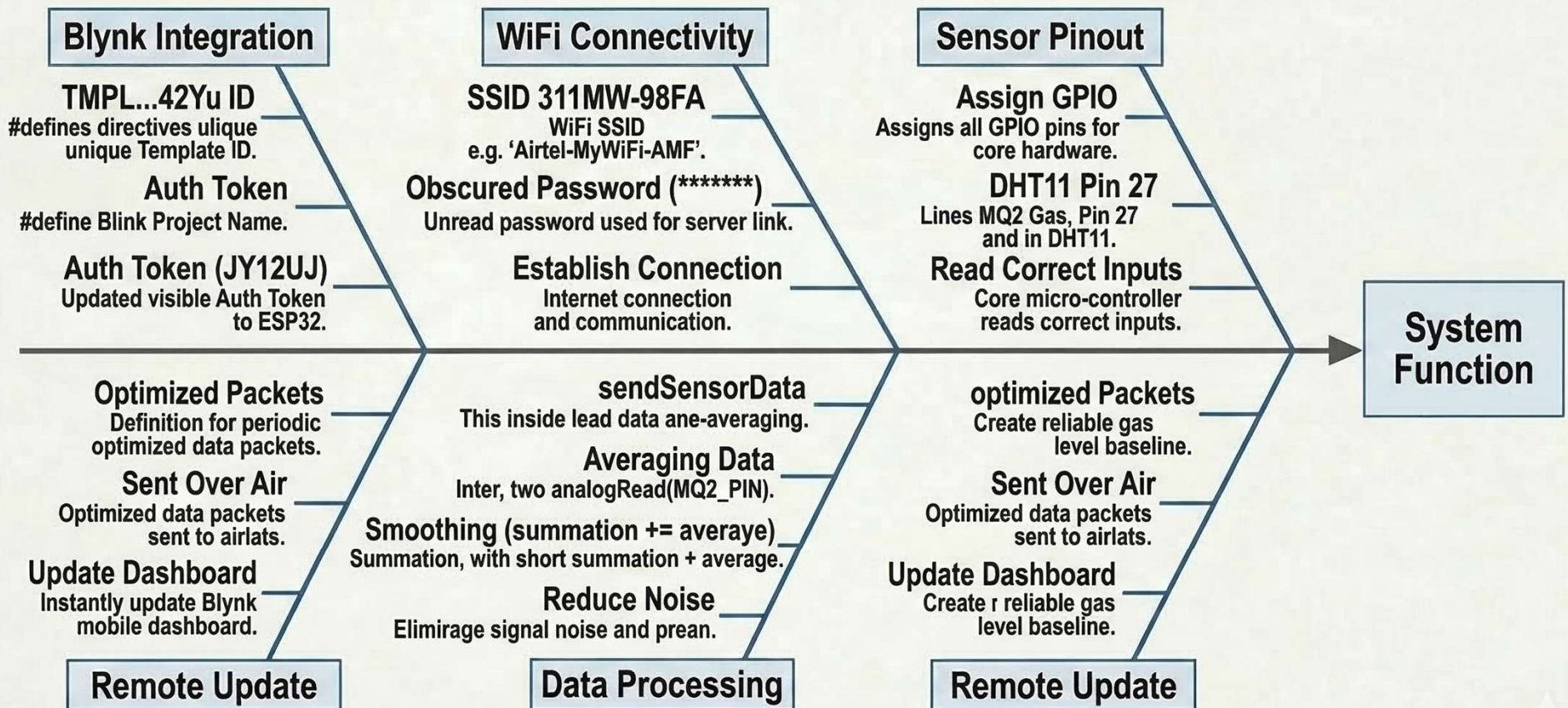
Sensor Pinout: This assigns all GPIO pins for all core hardware. Lines MQ2 Gas: Pin 27 and in DHT11. Core micro-controller sensors reads from the correct physical inputs.

Data Smoothing (sendSensorData function): This inside lead data ane-averaging reading inter, two analogrearrdroadt(si of (analogRead(MQ2_PIN)), with short, then summatin += averaye to elimirage signal noise and eliminate signal noise and prean, reliable gas level baseline.

Remote Dashboard Update: The defination for periodic, optimized data packets sent over the air and to those the arilats, instantly updating the Blynk mobile dashboard for immediate environmental monitoring.

```
1  #define BLYNK_TEMPLATE_ID "TMPL3vvu42Yu"
2  #define BLYNK_TEMPLATE_NAME "DRP PROJECT"
3  #define BLYNK_AUTH_TOKEN "lic3vdgyAuze75BsQRF4c6Bin-1YI3UJ"
4
5  #define BLYNK_PRINT Serial
6  #include <NiFI.h>
7  #include <NiFIClient.h>
8  #include <SlynkSimpleEsp32.h>
9  #include <DHT.h>
10
11 char ssid[] = "Airtel-MyWiFi-AMF-311MW-98FA";
12 char pass[] = "48eea83b";
13
14 #define MQ2_PIN 34
15 #define ALARM_PIN 14
16 #define DHTPIN 27
17 #define DHTTYPE DHT11
18
19 DHT dht(DHTPIN, DHTTYPE);
20 BlynkTimer timer;
21
22 // Adjust this based on your Serial Monitor "Clean Ain" reading
23 int sensorBaseline = 300;
24
25 void sendSensorData() {
26   // 1. Read Gas & Smooth
27   long sum = 0;
28   for(int i = 5; i < 50; i++) { sum += analogRead(MQ2_PIN); delay(2); }
```

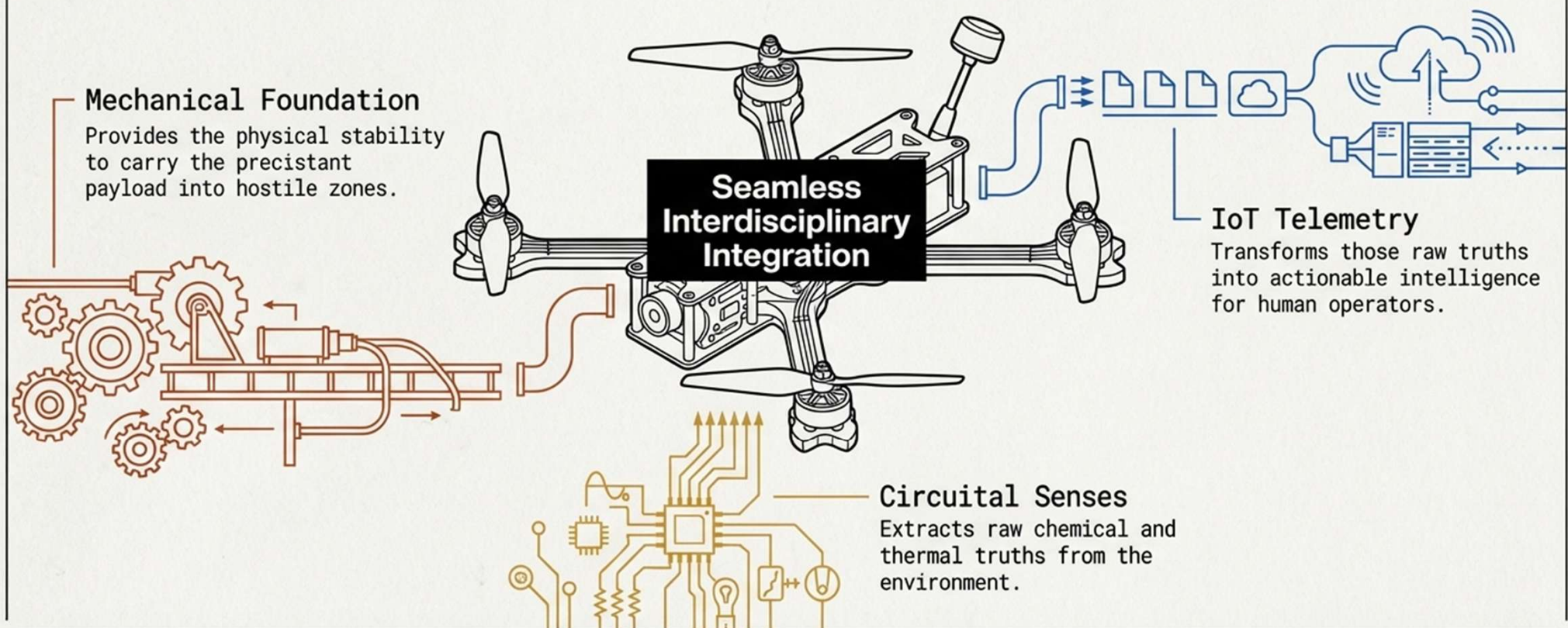
Fishbone Diagram



Diagnostic capabilities of the environmental payload.

Sensor Type Type	Target Variable	Output Format	Disaster Application
MQ-2 Gas Sensor	Combustible Gases (LPG, Smoke, Methane)	Analog (Requires ADC conversion)	Early-stage fire detection and toxic leak mapping.
DHT Sensor	Ambient Temperature & Relative Humidity	Digital Data Stream	Thermal profiling of disaster zones to predict fire spread.

System Convergence: The whole is greater than the sum of its parts.



A realized architecture for rapid disaster response.

By successfully unifying structural mechanics, environmental circuital arrays, and resilient IoT telemetry, this project transitions thretical engineering into a functional tool for crisis mitigation. The intelligence gathered by this platform replaces human risk with actionable data.



Time for the demo